25901411 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCMOS200 - WIRELESS COMMUNICATION TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. In cellular systems, FDMA separates users by assigning [K] / [CO1]

   a). Different time slots on the same frequency

   b). Different frequencies to different users

   c). Different power levels

   d). Different modulation schemes

2. Which multiple access technique divides users by frequency? [K] / [CO2]

   a). TDMA

   b). FDMA

   c). CDMA

   d). SDMA

3. Which hardware is commonly interfaced with GNU Radio? [K] / [CO3]

   a). Printer

   b). USRP

   c). Monitor

   d). Scanner

4. Digital modulation is preferred in modern systems due to [U] / [CO1]

a). Simple circuits b). Low cost

c). Better noise performance d). Low frequency use

5. In CDMA, each user is assigned [K] / [CO2]

a). A unique frequency b). A unique time slot

c). A unique spreading code d). A unique antenna

6. The Digital Signal Processor in SDR is used for [K] / [CO3]

a). Power amplification b). Frequency generation

c). Signal processing algorithms d). Antenna matching

7. Highest data capacity in wireless communication is achieved using [U] / [CO1]

a). LF band b). HF band

c). VHF band d). Microwave band

8. The Base Station Subsystem (BSS) plays a critical role in GSM by [U] / [CO2]

a). Handling billing and authentication b). Managing radio resources and handovers at the cell level

c). Maintaining subscriber databases d). Encrypting user data

9. SDR helps in reducing [U] / [CO3]

a). Antenna efficiency b). System adaptability

c). Development cost and time d). Signal processing speed

10. Which frequency band is commonly used in HAM radio? [K] / [CO3]

a). Optical band b). Microwave only

c). HF, VHF, and UHF bands d). Infrared band

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Explain free-space wireless communication and list the limitations associated with it. [U] / [CO1]

12. Describe the Bluetooth protocol stack. [U] / [CO2]

13. Explain the concept of blocks and flowgraphs in GNU Radio. [U] / [CO3]

14. Discuss the advantages and limitations of Wi-Fi technology. [U] / [CO2]

15. Discuss recent developments and applications of SDR technology. [An] / [CO3]

16. Explain the basic principle of operation of GPS. [K] / [CO2]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the basic principle of modulation and justify the need for modulation in wireless communication with respect to antenna height, interference, and multiplexing. [U] / [CO1]

18. Explain the need for modulation in communication systems. Discuss different types of noise and their effects on signal transmission. [U] / [CO2]

19. Explain the objectives and philosophy of GNU software. Discuss the importance of free and open-source software in communication systems. [U] / [CO3]

20. With neat block diagrams, explain AM and FM transmitters and highlight the key differences between them. [U] / [CO1]

[10x2 = 20]